TECHNICAL DOCUMENTATION / DOCUMENTACIÓN TÉCNICA

TORNEO GAMER DB

Bilingual project dossier for repository reviewers, instructors, and hiring audiences.

Dossier bilingüe para revisores del repositorio, docentes y audiencias técnicas.

Project / Proyecto	Oracle SQL and PL/SQL database for competitive eSports tournament management.
Repository focus / Enfoque	Schema design, seed data, analytical SQL, stored PL/SQL logic, and bilingual documentation.
Author of the system / Autor del sistema	pbenavides908
Document purpose / Propósito	Make the project understandable without opening every script individually.
Audience / Audiencia	Instructors, recruiters, peers, and technical reviewers.

Executive abstract / Resumen ejecutivo

This document explains what the project does, how the scripts are organized, how the data model works, which SQL and PL/SQL capabilities are demonstrated, and what future improvements are recommended. It is designed to let any reviewer understand the repository quickly and with confidence.

1. Purpose, audience, and review path

Final bilingual technical reference for the docs folder.

Propósito (ES)	Purpose (EN)
Este documento está pensado para acompañar el repositorio y reducir la fricción de revisión.	This document is meant to sit inside the repository and make technical review faster.
• Explica qué problema resuelve el sistema. • Resume la arquitectura de scripts. • Describe el modelo de datos y las reglas de negocio. • Muestra qué capacidades SQL y PL/SQL se evidencian. • Deja clara la ruta de mejora del proyecto.	• It explains the problem solved by the system. • It summarizes the script architecture. • It describes the data model and the main business rules. • It shows which SQL and PL/SQL capabilities are demonstrated. • It clarifies the future evolution path of the project.

Quick reference / Referencia rápida

Item	Value / Valor	Why it matters / Importancia
Project	Torneo Gamer DB	Clear technical identity for portfolio review
Database	Oracle	Matches course scope and PL/SQL focus
Author	pbenavides908	System ownership and authorship
Primary folder	/sql and /docs	Code plus explanatory material
Deliverables	Schema, data, queries, PL/SQL, PDF	Complete project package

2. Repository structure and execution order

The repository is intentionally organized in a review-friendly sequence.

Repository map / Mapa del repositorio

Path	Content	Reviewer expectation
/sql/01_schema.sql	Tables, keys, checks, and relationships	Database structure creation
/sql/02_data.sql	Twenty sample records per table	Consistent test data for demos
/sql/03_queries.sql	Analytical and operational SQL queries	Evidence of SQL command of joins, filters, ordering, aggregates
/sql/04_plsql.sql	Functions, procedures, cursors, blocks	Evidence of procedural database logic

/docs/documentation_bilingual.pdf	Bilingual documentation	Fast technical understanding without opening all scripts

Orden de ejecución (ES)	Execution order (EN)
1) Ejecutar 01_schema.sql. 2) Cargar 02_data.sql. 3) Probar 03_queries.sql. 4) Ejecutar 04_plsql.sql con SERVEROUTPUT activado. El orden es importante porque cada capa depende de la anterior.	1) Run 01_schema.sql. 2) Load 02_data.sql. 3) Test 03_queries.sql. 4) Run 04_plsql.sql with SERVEROUTPUT enabled. The order matters because each layer depends on the previous one.

3. Data model and business meaning

A compact relational model focused on competitive tournament management.

Visión del modelo (ES)	Model overview (EN)
El modelo está diseñado para representar un circuito competitivo sencillo pero realista. Cada tabla cumple una responsabilidad clara y evita duplicar información crítica. Se prioriza la legibilidad académica: nombres claros, relaciones explícitas y reglas de integridad fáciles de revisar.	The model is designed to represent a compact but realistic competitive circuit. Each table has a clear responsibility and avoids duplicating critical information. Academic readability is prioritized: clear names, explicit relationships, and integrity rules that are easy to review.

Entities / Entidades

Table	Core role	Key fields
jugadores	Stores player identities and profile data	jugador_id, nickname, correo, pais, estado
equipos	Stores teams and captain reference	equipo_id, nombre_equipo, tag_equipo, capitan_jugador_id
jugadores_equipos	Bridge between players and teams	jugador_id, equipo_id, rol, fecha_union, estado
torneos	Stores event definitions	torneo_id, juego, plataforma, premio_total, estado
partidas	Stores scheduled or completed matches	torneo_id, equipo_local_id, equipo_visitante_id, ganador_equipo_id
puntajes	Stores performance outcomes per	partida_id, equipo_id, puntos_obtenidos, kills,

	team in a match	resultado

Relationship logic / Lógica relacional

Source	Target	Type	Meaning
jugadores	equipos	N:M	A player may join teams through jugadores_equipos
equipos	jugadores	1:1 reference	Each team may declare one captain
torneos	partidas	1:N	One tournament can contain many matches
partidas	equipos	N:1 x2	Each match references local and visiting team
partidas	puntajes	1:N	A match can store score entries per team

4. Integrity rules and validation criteria

Constraints were selected to keep the model simple, demonstrable, and reliable.

Examples of implemented constraints / Ejemplos de restricciones implementadas

Category	Applied in	Purpose
Primary keys	All tables	Guarantee record uniqueness
Unique keys	nickname, correo, team names, tags	Prevent duplicate identities and labels
Check constraints	estado, género, modalidad, resultados	Restrict valid domain values
Foreign keys	Bridge, matches, scores	Preserve relational consistency
Identity columns	Main identifiers	Simplify inserts and improve demo readability

Controles de negocio (ES)	Business controls (EN)
• Los equipos no pueden enfrentarse a sí mismos en una partida.	• Teams cannot play against themselves in the same match.
• Los puntajes y métricas numéricas no pueden ser negativos.	• Scores and numeric metrics cannot be negative.
• Los estados se limitan a catálogos controlados.	• Status fields are restricted to controlled value sets.
• La tabla puente evita duplicados de combinación jugador-equipo.	• The bridge table avoids duplicate player-team combinations.
• El capitán del equipo se referencia desde la entidad jugadores.	• The team captain is referenced from the jugadores entity.

5. Sample data strategy

The seed layer is intentionally broad enough to power reporting and procedure tests.

Estrategia de carga (ES)	Seeding strategy (EN)
Se incorporan veinte registros por tabla principal para garantizar variedad suficiente en pruebas, consultas y procedimientos.	Twenty records are inserted per main table to provide enough variation for testing, querying, and procedure execution.
El objetivo no es simular producción, sino entregar un entorno consistente para demostrar relaciones, filtros, agregaciones y lógica procedural.	The purpose is not to mimic production volume, but to provide a consistent environment to demonstrate relationships, filtering, aggregation, and procedural logic.

Coverage provided by sample data / Cobertura entregada por los datos

Layer	Coverage	Why it helps
Players	Different countries, genders, and active statuses	Allows segmentation and identity checks
Teams	Distinct tags, rankings, and captains	Supports joins and reporting
Tournaments	Multiple games, platforms, states, and prize ranges	Supports categorization and status updates
Matches	Played and scheduled states	Supports result registration and pending views
Scores	Pairwise performance metrics	Supports ranking, sums, averages, and winner analysis

6. SQL query layer

A reviewer should quickly see both operational and analytical querying ability.

Qué demuestra la capa SQL (ES)	What the SQL layer demonstrates (EN)
La colección de consultas fue pensada para enseñar dominio de SELECT, JOIN, filtros, ORDER BY, GROUP BY y uso de TO_CHAR para salidas más profesionales. Se incluyen consultas operativas (listar, buscar, revisar estados) y consultas analíticas (ranking, promedios, conteos y acumulados).	The query collection was designed to show command of SELECT, JOIN, filtering, ORDER BY, GROUP BY, and TO_CHAR for more polished output. It includes operational queries (listing, searching, state checks) and analytical queries (rankings, averages, counts, and totals).

Representative SQL scenarios / Escenarios SQL representativos

Query family	Example objective	Skills demonstrated
Roster views	List players with teams and roles	Join chain, readable aliases
Tournament views	Show tournaments with formatted dates and prizes	TO_CHAR, ordering, presentation
Match monitoring	Identify played vs scheduled matches	State filtering, multi-join output
Performance analytics	Rank teams by points or kills	SUM, GROUP BY, ordering by aggregate
Portfolio-ready reports	Summarize wins, averages, and participation	Aggregations, left joins, null-safe logic

7. PL/SQL layer

Stored logic converts the repository from static schema to executable application behavior.

PL/SQL components / Componentes PL/SQL

Object	Purpose	Technical value
Anonymous blocks	Test lookups and print contextual output	Shows variable handling and exception branches
Functions	Return victories, total points, and prize categories	Shows reusable database logic
Procedures	Update states and register winners	Shows transaction-facing operations
Cursor-based	Print team ranking sequentially	Shows explicit cursor control

procedure		
DBMS_OUTPUT	Expose execution evidence	Improves demonstrability for class or review

Valor académico de la capa procedural (ES)	Academic value of the procedural layer (EN)
La lógica PL/SQL cumple dos objetivos: primero, demostrar que el autor no se queda solo en consultas; segundo, acercar el proyecto a un sistema real con operaciones reutilizables. Las funciones sirven para cálculo y clasificación. Los procedimientos sirven para cambios controlados sobre torneos y partidas.	The PL/SQL logic serves two goals: first, to show that the author goes beyond standalone queries; second, to move the repository closer to a real system with reusable operations. Functions are used for calculation and classification. Procedures are used for controlled changes over tournaments and matches.

8. Reviewer guide and expected evidence

What a teacher, recruiter, or peer should look at first.

Suggested review path / Ruta sugerida de revisión

Step	Open	What to verify
1	README + this PDF	Project purpose and structure are immediately understandable
2	01_schema.sql	Naming, relational consistency, constraints
3	02_data.sql	Data coherence and testing readiness
4	03_queries.sql	Breadth of SQL operations and readability
5	04_plsql.sql	Procedural logic, reuse, exception handling
6	Execution screenshots (optional)	Evidence that scripts ran successfully

Checklist de calidad (ES)	Quality checklist (EN)
• Nombres claros y consistentes. • Restricciones que responden a reglas del negocio. • Datos suficientemente variados para mostrar resultados significativos. • Consultas ordenadas y comentables en una demo. • Procedimientos y funciones con propósito concreto. • Documentación que permite entender el proyecto sin ayuda externa.	• Clear and consistent naming. • Constraints that map to business rules. • Data varied enough to produce meaningful outputs. • Queries that are easy to explain in a demo. • Procedures and functions with a concrete purpose. • Documentation that makes the project understandable without external help.

9. Expansion roadmap

Natural next steps if the repository is extended in future iterations.

Recommended improvements / Mejoras recomendadas

Improvement	Why it matters	Possible implementation
PACKAGE	Groups logic under a coherent API	pkg_torneo_gamer with functions and procedures
Transactions	Shows stronger control over failures	SAVEPOINT, ROLLBACK, business validations
Bulk processing	Shows performance-oriented techniques	BULK COLLECT and FORALL for batch updates
Triggers	Automates validations or audit	Result validation or change history

	trails	
Views / reports	Creates reviewer-friendly access points	Ranking view, active tournaments view
Security / roles	Adds realism to system governance	Read-only vs maintenance access patterns

10. Spanish quick reference

A concise index for Spanish-speaking reviewers.

Resumen ejecutivo en español

Tema	Resumen
Qué es	Una base de datos Oracle para gestionar torneos gamer con enfoque académico y de portafolio.
Qué contiene	Estructura relacional, carga de datos, consultas SQL, procedimientos y funciones PL/SQL, y documentación bilingüe.
Qué demuestra	Dominio de modelado relacional, consultas, restricciones, lógica procedural y presentación técnica.
Autor del sistema	pbenavides908
Uso recomendado	Portafolio estudiantil, evaluación docente, revisión técnica inicial.

11. English quick reference

A concise index for English-speaking reviewers.

English executive index

Topic	Summary
What it is	An Oracle database project for managing gamer tournaments with both academic and portfolio value.
What it contains	Relational structure, seed data, SQL queries, PL/SQL procedures and functions, and bilingual documentation.
What it demonstrates	Relational modeling, querying, constraints, procedural logic, and technical presentation.
System author	pbenavides908
Recommended use	Student portfolio, instructor assessment, initial technical review.

12. Documentation control

Basic metadata for versioning and maintenance.

Document control / Control documental

Field	Value
Document title	Torneo Gamer DB - Bilingual Technical Documentation
Version	1.1 - visual redesign
Owner / Author of the system	pbenavides908
Language scope	Spanish and English
Intended folder	/docs
Related artifacts	01_schema.sql, 02_data.sql, 03_queries.sql, 04_plsql.sql, README.md

Closing note / Nota de cierre

This PDF is intentionally more detailed than a README. Its role is to help a reviewer understand the system, the repository flow, and the technical maturity of the project in a single reading pass.